

ASSURANCE OF LEARNING REPORT

COMPUTER NETWORKS



TITLE
NETWORKING FLOOR 2,4,5

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Computer Networks
Semester Ganjil 2024/2025

Report of Assurance of Learning Computer Networks

Title

NETWORKING FLOOR 2,4,5

TCP/IP LAYER

- **Layer 1 - Physical Layer**

Devices and Prices

Cat6a/UTP Cable = Rp 3.900/m

- **2nd Floor**

There are several key rooms on the second floor, including classrooms, the Library, BINUS Radio, BINUS TV, the Photography Studio, and the Greenscreen Studio. Each room is equipped with network connectivity to support devices such as computers and access points, ensuring smooth operation for academic and multimedia activities.

The placement of these connections has been strategically planned based on the layout of the floor to maximize efficiency and coverage. For example, the Library requires the highest bandwidth, with 15 connections, while other rooms such as BINUS Radio and BINUS TV have one connection each.

The chosen cable type for this floor is Cat6a, a high-performance twisted-pair cable known for its superior data transfer speeds and reduced interference. This cable is ideal for high-bandwidth environments such as the Library and multimedia studios, providing reliable connections for devices. Cat6a is also durable and readily available, making it both cost-effective and practical for installation on this floor.

A total of approximately 216 meters of Cat6a cable is required to connect all the rooms, with the Library using 80 meters, BINUS Radio 50 meters, BINUS TV 40 meters, and other classrooms and studios requiring between 30 and 50 meters each. For Rp. 3,900 per meter, the total expense for the cable is estimated to be **Rp. 842,400.**

- **4th Floor**

On the fourth floor, there are several key rooms, including the Auditorium, IT Room, LSC, and classrooms such as B0402, B0403, and C0410. Each room requires network connectivity to support devices like computers and access points, ensuring seamless operations for academic and administrative activities.

The placement of these connections has been carefully designed to maximize coverage and efficiency across the floor. For example, the Auditorium, which is a larger room, requires longer cabling compared to smaller classrooms.

The chosen cable type for the fourth floor is Cat6a, known for its high-performance capabilities and durability. This cable is ideal for environments with high data transfer requirements, such as the Auditorium and IT Room.

A total of approximately 228 meters of Cat6a cable is needed to connect all rooms. Key connections include 30 meters for the LSC, 18 meters for the IT Room, and 25 meters for C0412. At a cost of Rp. 3,900 per meter, the total expense for the cabling amounts to **Rp. 889,200.**

- **5th Floor**

On the fifth floor, the key rooms include the Auditorium, IT Room, ASA0501, ASA0503, ASA0504, ASA0505, C0509, and C0511. These rooms require network connectivity to support computers, access points, and other devices, ensuring smooth operation for academic and administrative activities. The placement of network connections was carefully planned based on the layout of the floor to optimize coverage and efficiency.

The cable used for this floor is Cat6a, chosen for its ability to handle high-speed data transfer and its durability. A total of approximately 217.5 meters of cable is required to connect all rooms. Major connections include 30 meters for C0511, 27 meters for the IT Room, and 18 meters for ASA0504. For Rp. 3,900 per meter, the total expense for the cabling is estimated to be **Rp. 848,250.**

- **Layer 2 - Data Link**

Devices and Price :

LINKSYS LGS108-AP 8-Port Business Desktop Gigabit Switch LGS108AP = Rp 697.000

- **2nd Floor**

On the second floor, there are three switches installed to support the network infrastructure across multiple rooms. These include key areas such as the Library, BINUS TV, BINUS Radio, the Photography Studio, and classrooms like C0209, C0210, and others. The LINKSYS LGS108-AP 8-Port Business Desktop Gigabit Switch is used, with each switch capable of handling up to eight devices.

The strategic placement of the three switches ensures that all rooms are connected efficiently, with minimal cable usage and optimal network performance. At a price of **Rp 697.000** per switch, the total cost for the second floor is **Rp 2.091.000**.

- **4th Floor**

On the fourth floor, there are four switches required to connect all the rooms, including the Auditorium, IT Room, LSC, and classrooms such as C0401, C0403, and C0410. The switches are placed strategically to ensure robust connectivity throughout the floor, supporting devices in high-traffic areas like the Auditorium and IT Room. Each switch, with its 8-port capacity, provides sufficient connections for the designated devices in its area. With each switch costing **Rp 697.000**, the total expense for the fourth floor is **Rp 2.788.000**.

- **5th Floor**

On the fifth floor, four switches are also used to connect key rooms such as the Auditorium, ASA0501, ASA0503, and ASA0504, and classrooms like C0509 and C0511. These switches are evenly distributed across the floor to handle the connectivity needs efficiently, ensuring stable performance even in high-demand areas like the Auditorium. Each LINKSYS LGS108-AP 8-Port Business Desktop Gigabit Switch supports multiple devices, making it an effective solution for this floor. The total cost for these four switches is **Rp 2.788.000**.

- **Layer 3 - Network Layer**

Devices and Prices:

Tp-Link Archer AX12 AX1500 Wi-Fi 6 Router Tp-Link Archer AX12 = Rp520.000

- **2nd Floor**

On the second floor, two routers are installed to ensure comprehensive Wi-Fi coverage across the floor. These routers are placed strategically, with one positioned near the central area to serve high-traffic spaces such as the Library and multimedia rooms (BINUS TV and BINUS Radio), and another located near the classrooms to cover C0209, C0210, and adjacent rooms. The routers used are Tp-Link Archer AX12 AX1500 Wi-Fi 6 Routers, known for their advanced performance and ability to handle multiple simultaneous connections with minimal latency. With each router costing Rp 520,000, the total cost for the two routers on this floor amounts to **Rp 1.040.000**.

- **4th Floor**

On the fourth floor, two routers are also deployed to ensure stable and efficient Wi-Fi connectivity throughout the floor. One router is positioned near the Auditorium to manage high-demand traffic, while the other is placed near the IT Room and adjacent classrooms to provide reliable coverage.

The routers' ability to support Wi-Fi 6 technology ensures high-speed data transfer and a seamless network experience for all connected devices. With each Tp-Link Archer AX12 AX1500 Router priced at Rp 520,000, the total expense for the fourth floor is **Rp 1.040.000**.

- **5th Floor**

On the fifth floor, two routers are placed to provide robust wireless connectivity to key rooms, including the Auditorium, ASA0501, and ASA0503, and classrooms like C0511 and C0509.

One router is strategically located near the Auditorium and adjacent rooms, while the second router covers the opposite wing of the floor. This setup ensures strong and consistent Wi-Fi signals across all areas. Using the same Tp-Link Archer AX12 AX1500 Wi-Fi 6 Routers, the total cost for the fifth floor is **Rp 1.040.000**

Floor 2:

- **Switch 1 (B0201, Library)**

- Usable IP : 192.168.0.1 - 192.168.1.254 /23
- Network Address : 192.168.0.0
- Broadcast Address : 192.168.1.255

- **Switch 0 (C0201, C0203, C0204, C0205, C0206, C0207, C0209, C0210)**

- Usable IP : 192.168.2.1 - 192.168.3.254 /23
- Network Address : 192.168.2.1
- Broadcast Address : 192.168.3.255

- **Switch 2 (Greenscreen, Studio Photography, BINUS TV, BINUS Radio, Amphitheater, C0212)**

- Usable IP : 192.168.4.1 - 192.168.6.254 /23
- Network Address : 192.168.4.0
- Broadcast Address : 192.168.6.255

Floor 4:

- **Switch 5 (Auditorium B0401, B0402, B0403, B0404, B0405)**
 - Usable IP : 192.168.8.1 - 192.168.9.254 /23
 - Network Address : 192.168.8.0
 - Broadcast Address : 192.168.9.255
- **Switch 4 (IT)**
 - Usable IP : 192.168.10.1 - 192.168.10.126 /25
 - Network Address : 192.168.10.0
 - Broadcast Address : 192.168.10.127
- **Switch 3 (C0411, C0412)**
 - Usable IP : 192.168.12.1 - 192.168.12.126 /25
 - Network Address : 192.168.12.0
 - Broadcast Address : 192.168.12.127
- **Switch 6 (LSC 0407, LSC 0408-09, C0401, C0403, C0406, C0410)**
 - Usable IP : 192.168.13.1 - 192.168.14.254 /23
 - Network Address : 192.168.13.0
 - Broadcast Address : 192.168.14.255

Floor 5:

- **Switch 8 (Auditorium, B0503, B0504)**
 - Usable IP : 192.168.16.1 - 192.168.16.254 /24
 - Network Address : 192.168.16.0
 - Broadcast Address : 192.168.16.255
- **Switch 7 (ASA 0501, ASA 0503, ASA 0504-05, ASA 0506)**
 - Usable IP : 192.168.17.1 - 192.168.18.254 /23
 - Network Address : 192.168.17.0
 - Broadcast Address : 192.168.18.255
- **Switch 9 (C0507, C0508, C0509)**
 - Usable IP : 192.168.19.1 - 192.168.19.254 /24
 - Network Address : 192.168.19.0
 - Broadcast Address : 192.168.19.255
- **Switch 10 (C0511)**
 - Usable IP : 192.168.20.1 - 192.168.20.62 /26
 - Network Address : 192.168.20.0
 - Broadcast Address : 192.168.20.63

Floor 2	Host	IP Range	Usable IP Range	Network Address	Broadcast Address	Subnet Mask
Library	250	192.168.0.0	192.168.0.1 - 192.168.1.254	192.168.0. 0	192.168.1.255	255.255.254.0 /23

B0201	250	192.168.1.0	192.168.0.1 - 192.168.1.254	192.168.0. 0	192.168.1.255	255.255.254.0 /23
Switch 1	500	192.168.0.0	192.168.0.1 -			
C0201	60	192.168.2.0	192.168.2.1 - 192.168.3.254	192.168.2. 1	192.168.3.255	255.255.254.0 /23
C0203	60	192.168.2.64	192.168.2.1 - 192.168.3.254	192.168.2. 1	192.168.3.255	255.255.254.0 /23
C0204	60	192.168.2.128	192.168.2.1 - 192.168.3.254	192.168.2. 1	192.168.3.255	255.255.254.0 /23
C0205	60	192.168.2.192	192.168.2.1 - 192.168.3.254	192.168.2. 1	192.168.3.255	255.255.254.0 /23
C0206	60	192.168.3.0	192.168.2.1 - 192.168.3.254	192.168.2. 1	192.168.3.255	255.255.254.0 /23
C0207	60	192.168.3.64	192.168.2.1 - 192.168.3.254	192.168.2. 1	192.168.3.255	255.255.254.0 /23
C0209	60	192.168.3.128	192.168.2.1 - 192.168.3.254	192.168.2. 1	192.168.3.255	255.255.254.0 /23
C0210	60	192.168.3.192	192.168.2.1 - 192.168.3.254	192.168.2. 1	192.168.3.255	255.255.254.0 /23
C0212	60	192.168.4.0	192.168.4.1 - 192.168.6.254	192.168.4. 0	192.168.6.255.	255.255.254.0 /23
Binus Radio	60	192.168.4.64	192.168.4.1 - 192.168.6.254	192.168.4. 64	192.168.6.255.	255.255.254.0 /23
Binus TV	60	192.168.4.128	192.168.4.1 - 192.168.6.254	192.168.4. 128	192.168.6.255.	255.255.254.0 /23
Photogra phy Studio	60	192.168.4.192	192.168.4.1 - 192.168.6.254	192.168.4. 192	192.168.6.255.	255.255.254.0 /23
Greenscr een	60	192.168.5.0	192.168.4.1 - 192.168.6.254	192.168.5. 0	192.168.6.255.	255.255.254.0 /23
Amphithe ater	60	192.168.5.64	192.168.4.1 - 192.168.6.254	192.168.5. 64	192.168.6.255.	255.255.254.0 /23

Floor 4	Host	IP Range	Usable IP Range	Network Address	Broadcast Address	Subnet Mask
Auditorium	250	192.168.8.0	192.168.8.1 - 192.168.9.254	192.168.8.0	192.168.9.255	255.255.254.0 /23
LSC C0408-09	250	192.168.9.0	192.168.13.1 - 192.168.14.254	192.168.13.0	192.168.14.255	255.255.254.0 /23
IT	120	192.168.10.0	192.168.10.1 - 192.168.10.126	192.168.10.0	192.168.10.127	255.255.255.128 /25
B0402	60	192.168.10.128	192.168.8.1 - 192.168.9.254	192.168.8.0	192.168.9.255	255.255.254.0 /23
B0403	60	192.168.10.192	192.168.8.1 - 192.168.9.254	192.168.8.0	192.168.9.255	255.255.254.0 /23
B0404	60	192.168.11.0	192.168.8.1 - 192.168.9.254	192.168.8.0	192.168.9.255	255.255.254.0 /23
B0405	60	192.168.11.64	192.168.8.1 - 192.168.9.254	192.168.8.0	192.168.9.255	255.255.254.0 /23
C0401	60	192.168.11.128	192.168.13.1 - 192.168.14.254	192.168.13.0	192.168.14.255	255.255.255.192 /26
C0403	60	192.168.11.192	192.168.13.1 - 192.168.14.254	192.168.13.0	192.168.14.255	255.255.254.0 /23
C0406	60	192.168.12.0	192.168.13.1 - 192.168.14.254	192.168.13.0	192.168.14.255	255.255.254.0 /23
C04010	60	192.168.12.64	192.168.13.1 - 192.168.14.254	192.168.13.0	192.168.14.255	255.255.254.0 /23
C0411	60	192.168.12.128	192.168.12.1 - 192.168.12.126	192.168.12.0	192.168.12.127	255.255.255.128 /25
C0412	60	192.168.12.192	192.168.12.1 - 192.168.12.126	192.168.12.0	192.168.12.127	255.255.255.128 /25
LSC C0407	60	192.168.13.0	192.168.13.1 - 192.168.14.254	192.168.13.0	192.168.14.255	255.255.254.0 /23

Floor 5	Host	IP Range	Usable IP Range	Network Address	Broadcast Address	Subnet Mask
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Auditorium	250	192.168.16.0	192.168.16.1 - 192.168.16.254	192.168.16.0	192.168.16.255	255.255.255.0 /24
ASA 0504-05	250	192.168.17.0	192.168.17.1 - 192.168.18.254	192.168.17.0	192.168.18.255	255.255.255.0 /24
ASA 0501	60	192.168.18.0	192.168.17.1 - 192.168.18.254	192.168.17.0	192.168.18.255	255.255.254.0 /23
ASA 0503	60	192.168.18.64	192.168.17.1 - 192.168.18.254	192.168.17.0	192.168.18.255	255.255.254.0 /23
ASA 0506	60	192.168.18.128	192.168.17.1 - 192.168.18.254	192.168.17.0	192.168.18.255	255.255.254.0 /23
B0503	60	192.168.18.192	192.168.16.1 - 192.168.16.254	192.168.16.0	192.168.16.255	255.255.255.0 /24
B0504	60	192.168.19.0	192.168.16.1 - 192.168.16.254	192.168.16.0	192.168.16.255	255.255.255.0 /24
C0507	60	192.168.19.64	192.168.19.1 - 192.168.19.254	192.168.19.0	192.168.19.255	255.255.255.0 /24
C0508	60	192.168.19.128	192.168.19.1 - 192.168.19.254	192.168.19.0	192.168.19.255	255.255.255.0 /24
C0509	60	192.168.19.192	192.168.19.1 - 192.168.19.254	192.168.19.0	192.168.19.255	255.255.255.0 /24
C0511	60	192.168.20.0	: 192.168.20.1 - 192.168.20.62	192.168.20.0	192.168.20.63	255.255.255.192 /26

Floor 2:

1. Next hop floor 4: 192.168.100.162
2. Next hop floor 5: 192.168.100.34

Floor 4:

1. Next hop floor 2: 192.168.100.161
2. Next hop floor 5: 192.168.100.145

Floor 5:

1. Next hop floor 2: 192.168.100.33
2. Next hop floor 4: 192.168.100.146

● Layer 4 - Transport Layer

Layer 4 is the transport layer which is responsible for establishing communication between application processes that are being executed on separate hosts. Every connection online is possible because of the protocols provided by the transport layer and there are many protocols available for this.

The most common ones are TCP and UDP. Every application can communicate via TCP or UDP protocols, both providing different ways with different pros and cons. There are several services provided by the transport layer such as:

1. Error control
2. Sequence control
3. Loss Control
4. Duplication control
5. Flow control
6. Multiplexing
7. Etc

For the protocols, TCP and UDP have different ways for applications to communicate. TCP is a connection-based protocol, it can sequence data and retransmit data if packets fail to arrive, ensuring the data is delivered. While it is slower than UDP, TCP will ensure the data is fully delivered. TCP uses a 3-way handshake before the communication process starts so that both ends can initiate, negotiate, and separate TCP socket connections. TCP is best for transferring data such as text messages, files, etc. that must be fully delivered.

On the other hand, UDP is a connectionless based protocol, it doesn't establish any connection when sending data. UDP sends data without confirming receipt or checking for errors and packet losses. Therefore, with UDP full data delivery is not guaranteed. UDP prioritizes fast data delivery over complete data delivery. UDP is best for online gaming, live streaming, etc.

TCP and UDP facilitate communication between the network layer and application layer protocols such as HTTP, FTP, and POP. To manage multiple processes and ensure data is delivered to the correct application, the transport layer uses port numbers contained in its headers.

Port numbers range from 0 to 65,535 and are divided into three categories:

1. Well-Known Ports (0–1023):

- Reserved for widely-used protocols and services.
- Examples include HTTP (80), HTTPS (443), FTP (20, 21), SMTP (25), and DNS (53).
- These ports are typically used for system-level services and require administrative privileges to configure.

2. Registered Ports (1024–49151):

- Assigned to user applications or specific processes that require network connectivity.

- Applications, such as those for authentication or communication services, can request ports in this range.

3. Dynamic or Private Ports (49152–65535):

- Assigned dynamically to client applications when a connection is initiated.
- These ports are used temporarily and are not tied to specific services, making them suitable for short-term communications.

For Binusmaya-specific activities, such as NIM authentication, ports can be registered in the *Registered Ports* range to provide dedicated communication channels. Standard activities, such as HTTPS, FTP, and SMTP, will utilize the *Well-Known Ports* range, while the *Dynamic Port* range can handle other client-side connections.

A socket address is created by combining an IP address with a port number. This uniquely identifies processes on a network. The client's socket address identifies the client process, while the server's socket address identifies the server process, ensuring data is routed correctly between them.

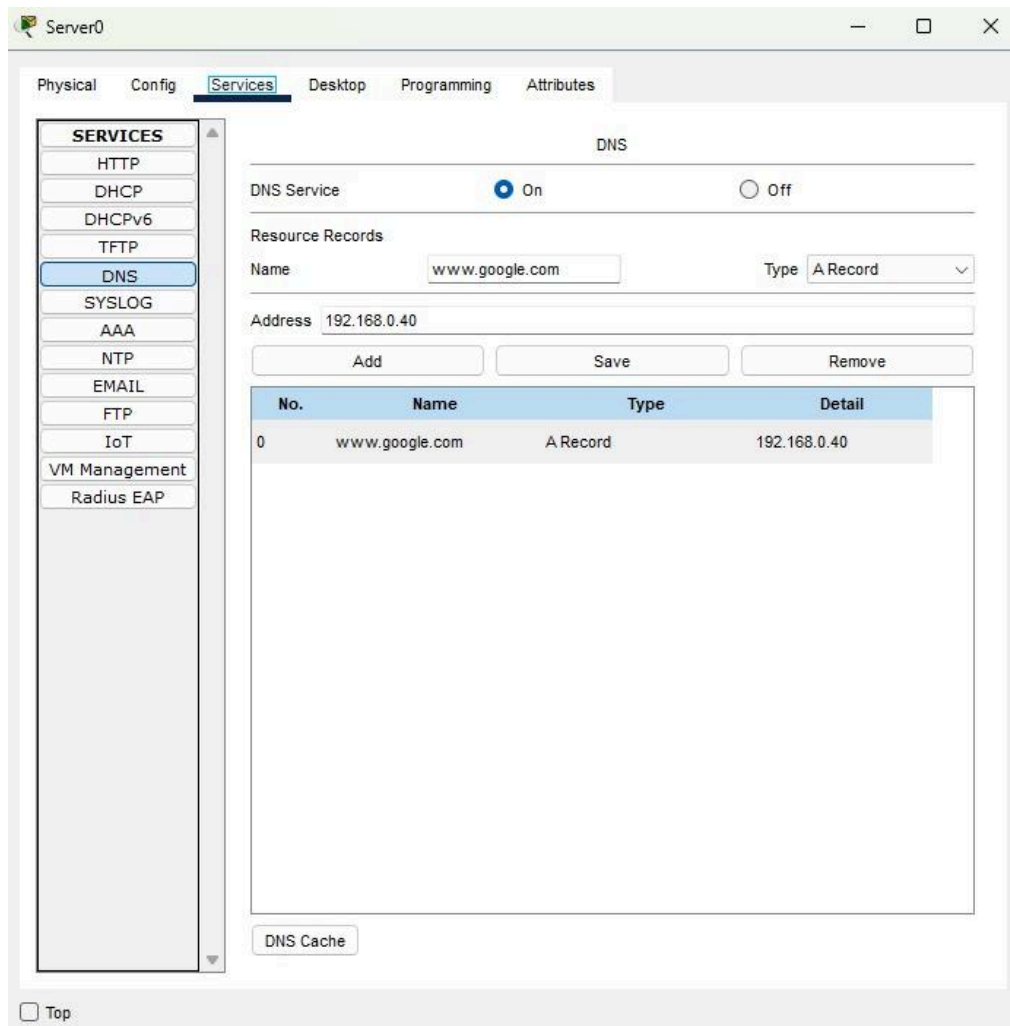
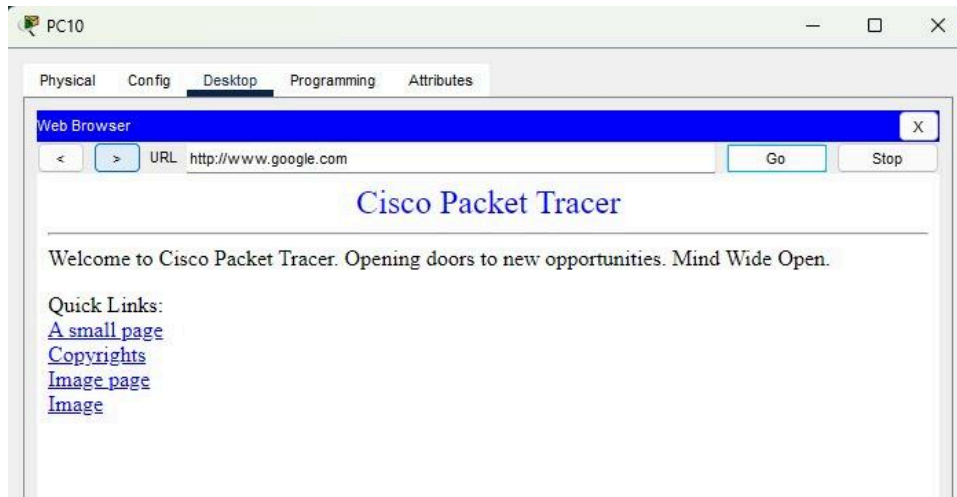
Understanding and assigning appropriate port numbers is critical for enabling seamless communication across network layers

• Layer 5 - Application Layer

Domain Name System (DNS)

Domain Name System (DNS) translates domain names to readable IP addresses for the computer. The presence of DNS servers eliminates the need for humans to memorize IP addresses such as 192.168.0.40, instead they can just type in their destination domain such as google.com.

We have established a DNS server for our AOL on the third floor, linked to the library room's switch. Despite its physical location on the third floor, it is accessible by all floors – third, fifth, and sixth. The reason for this is because one server can accommodate all the request needs of the client.



Simple Mail Transfer Protocol (SMTP)

Simple Mail Transfer Protocol (SMTP) can be defined as a transport protocol used by mail servers to send, receive, and relay outgoing emails between senders and receivers. SMTP is built on top of TCP to ensure reliable delivery of mail packets.

When a user sends an email, the server receives information such as the sender's and receiver's email addresses and the content of the email. After receiving the sender's and receiver's email addresses, the MTA (Mail Transfer Agent) checks whether both email addresses belong to the same domain.

If the email addresses are in the same domain, the 19 emails will be sent immediately. However, if they are not in the same domain, the server will use DNS to identify the recipient's domain and send the email to the correct server. We placed the SMTP server on the third floor linked to the library and configured it for five users. To test the SMTP functionality, we attempted to send an email with the subject "hello (name)" from one PC to another, using different user accounts.

PC8

Physical Config Desktop Programming Attributes

Configure Mail X

User Information

Your Name: delon

Email Address: delon@gmail.com

Server Information

Incoming Mail Server: 192.168.0.40

Outgoing Mail Server: 192.168.0.40

Logon Information

User Name: delon

Password:

Save Remove Clear Reset

Top

PC7

Physical Config **Desktop** Programming Attributes

Configure Mail X

User Information

Your Name: wilbert

Email Address: wilbert@gmail.com

Server Information

Incoming Mail Server: 192.168.0.40

Outgoing Mail Server: 192.168.0.40

Logon Information

User Name: wilbert

Password:

Save Remove Clear Reset

☐ Top

Server0

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL**
- FTP
- IoT
- VM Management
- Radius EAP

EMAIL

SMTP Service ☒ ON ☐ OFF

POP3 Service ☒ ON ☐ OFF

Domain Name: gmail.com Set

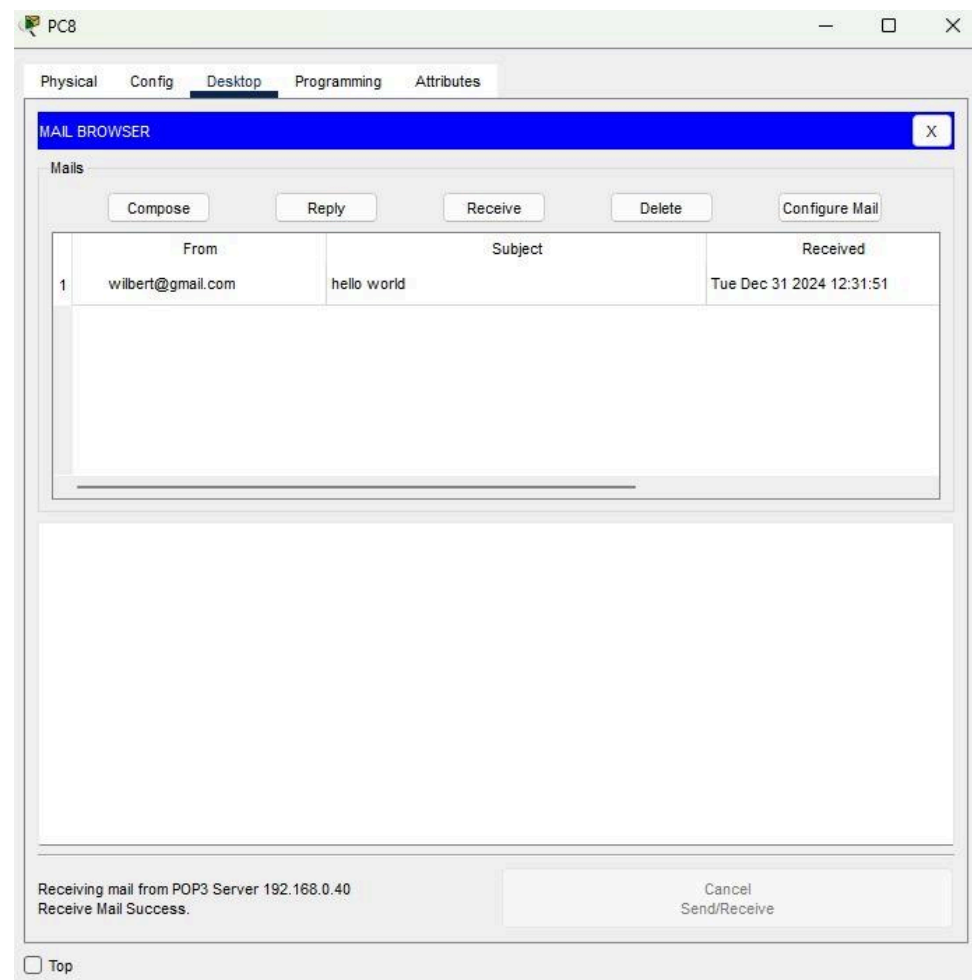
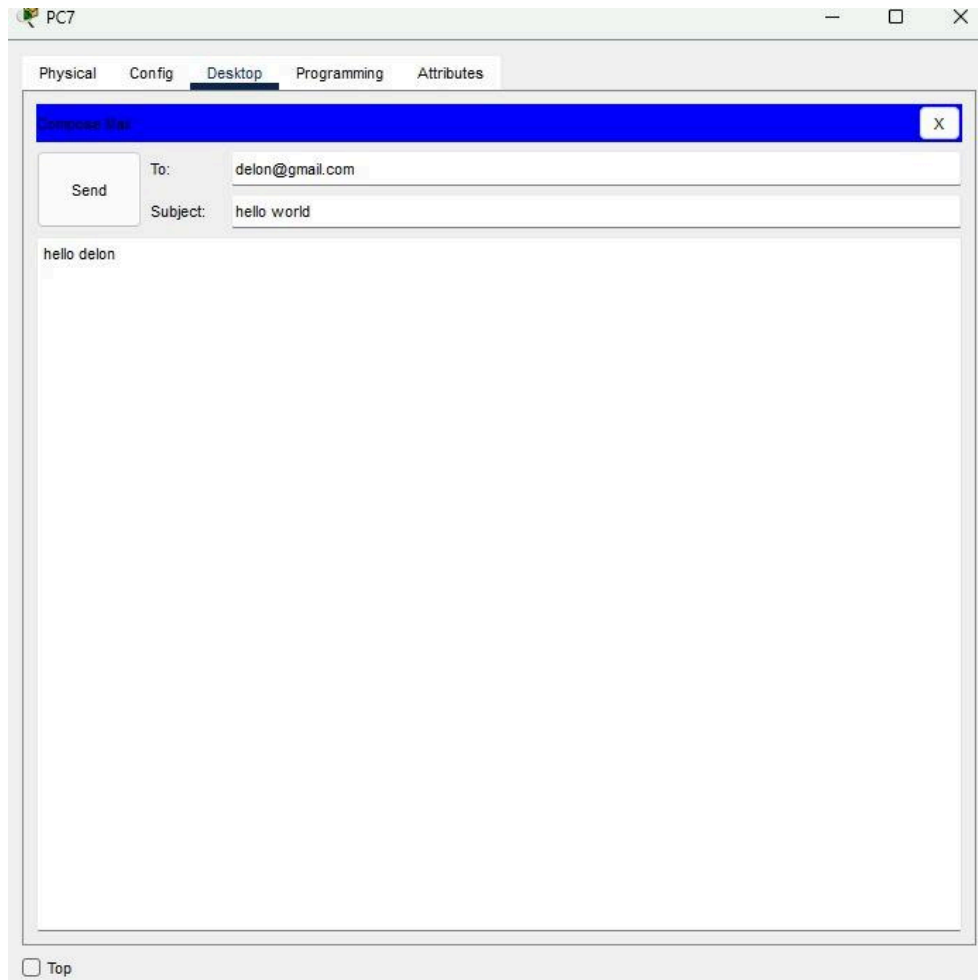
User Setup

User Password

wilbert
delon

+
-
Change
Password

☐ Top

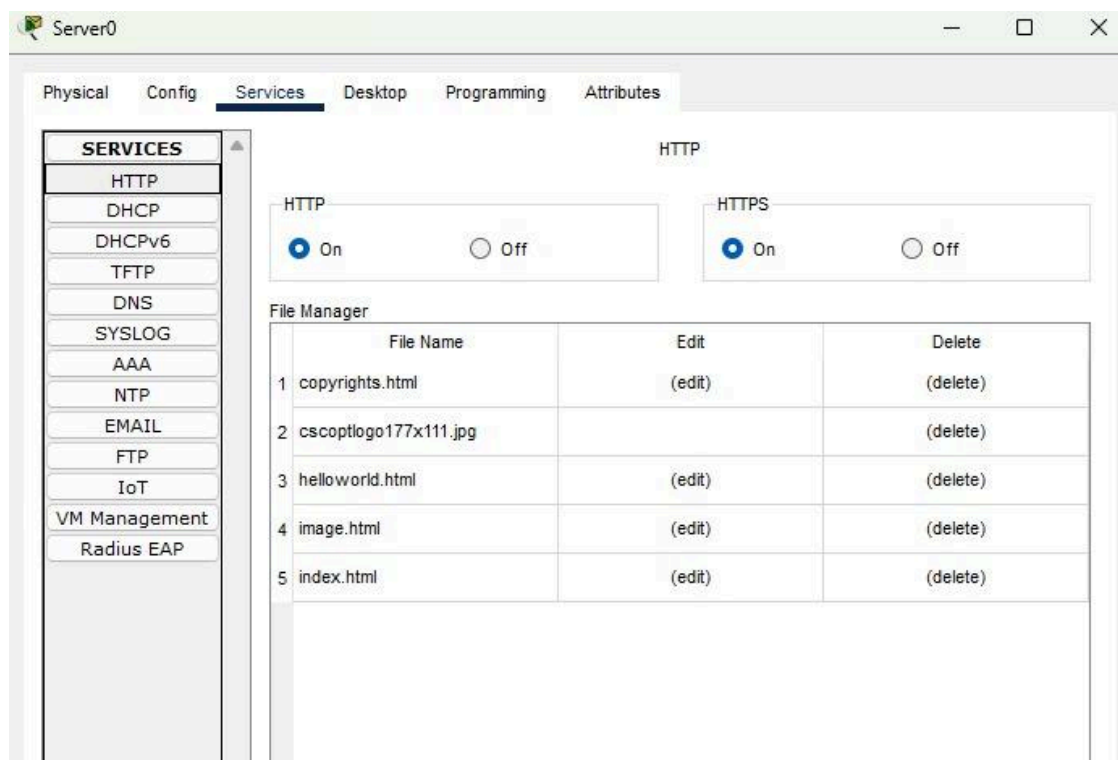
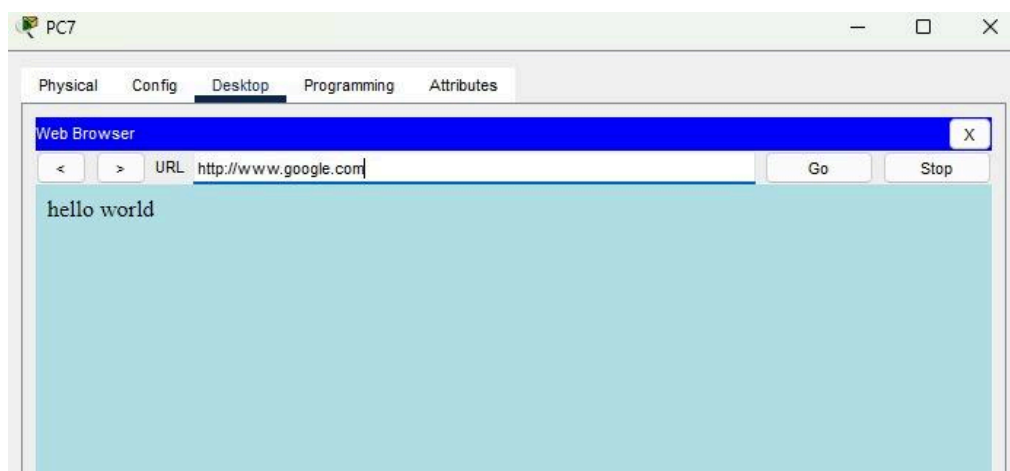


Hypertext Transfer Protocol (HTTP)

Hypertext Transfer Protocol (HTTP) is an application protocol that loads web pages using hypertext links. Essentially, HTTP is a protocol that transfers data between devices that are connected to the internet. HTTP is divided into HTTP and HTTPS, which is HTTP with encryption and verification.

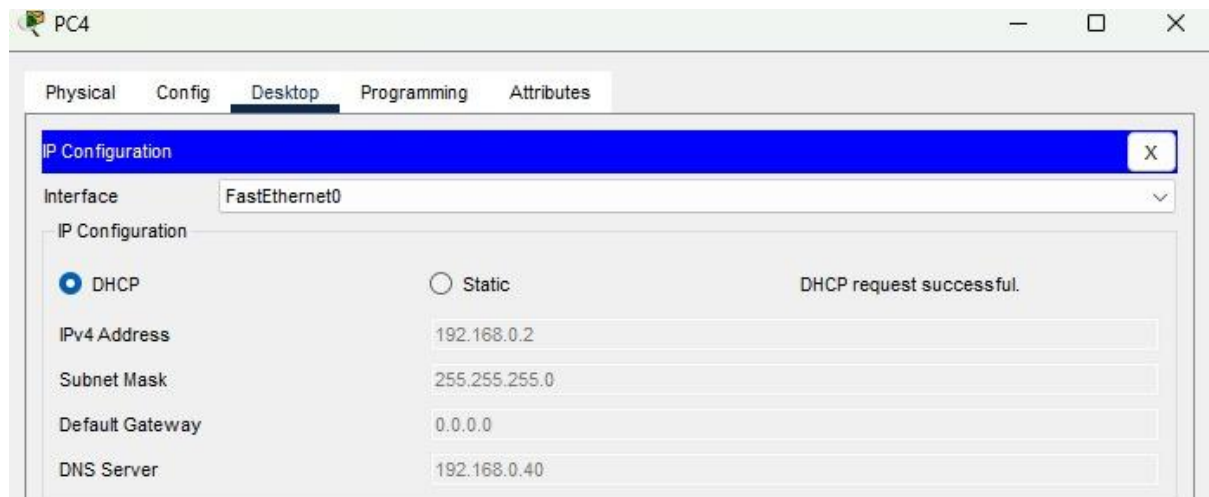
In short, HTTPS is more secure than the original HTTP since it uses TLS (Transfer Layer Security) to encrypt normal HTTP. We turned on access to both HTTP and HTTPS in Server0 in the library room since it increases the number of web pages accessible by the computer.

An example of a normal HTTP website can be seen on the 2nd image, <https://www.google.com> which is accessible through the Cisco Packet Tracer. In the 2nd image, you can see the HTML page with its CSS – the blue background color and different text styling – which indicates that it's an actual web structure and that the HTTP is working.



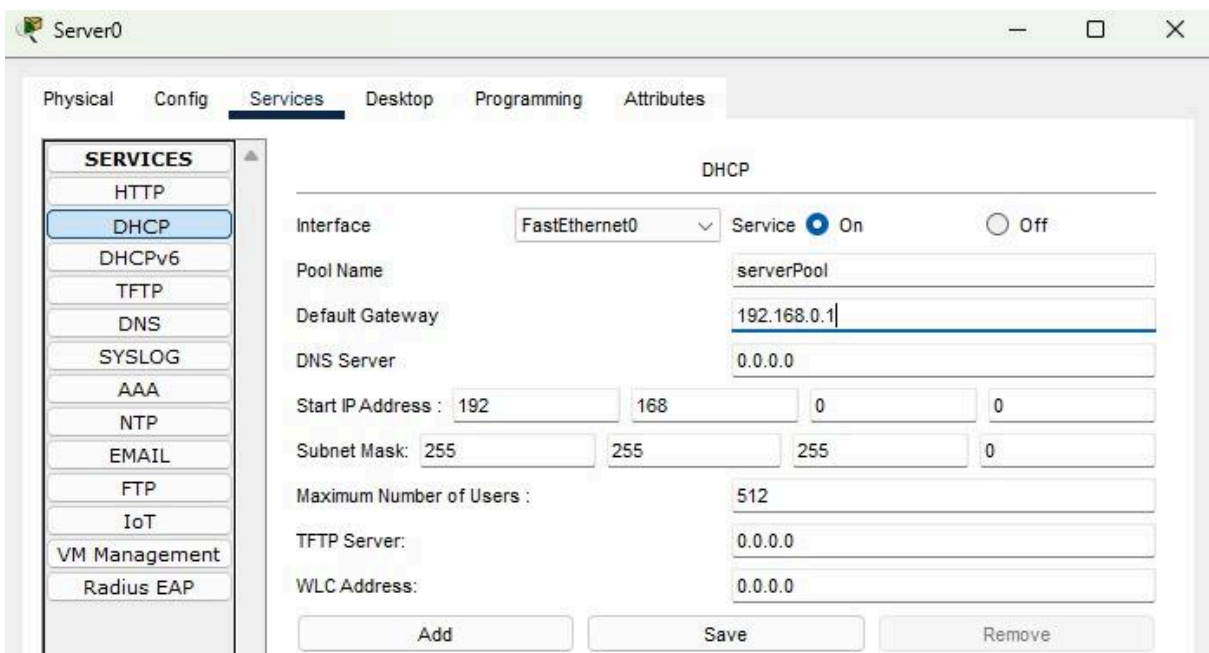
Dynamic Host Configuration Protocol (DHCP)

Dynamic Host Configuration Protocol (DHCP) is a network protocol that automatically assigns IP addresses and other network settings to devices on a network. By using DHCP, it takes the hassle out of manually assigning IP addresses for every device on a network.



The screenshot shows the 'IP Configuration' window for 'PC4'. The 'Desktop' tab is selected. The 'Interface' is 'FastEthernet0'. Under 'IP Configuration', the 'DHCP' radio button is selected, and the 'Static' radio button is unselected. A message 'DHCP request successful.' is displayed. The following fields are filled:

Field	Value
IPv4 Address	192.168.0.2
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	192.168.0.40



The screenshot shows the 'DHCP' configuration window for 'Server0'. The 'Services' tab is selected. The 'Interface' is 'FastEthernet0'. The 'Service' is 'On'. The 'Pool Name' is 'serverPool'. The 'Default Gateway' is '192.168.0.1'. The 'DNS Server' is '0.0.0.0'. The 'Start IP Address' is '192.168.0.0'. The 'Subnet Mask' is '255.255.255.0'. The 'Maximum Number of Users' is '512'. The 'TFTP Server' is '0.0.0.0'. The 'WLC Address' is '0.0.0.0'. The 'Add', 'Save', and 'Remove' buttons are visible at the bottom.

Field	Value
Interface	FastEthernet0
Service	On
Pool Name	serverPool
Default Gateway	192.168.0.1
DNS Server	0.0.0.0
Start IP Address	192.168.0.0
Subnet Mask	255.255.255.0
Maximum Number of Users	512
TFTP Server	0.0.0.0
WLC Address	0.0.0.0

Group Leader

Lecturer

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